

OFFCODER

Local-First Autonomous Inference Offload & Verified Coding Harness

SPEC // PRODUCTION

\$0.00

API INGRESS COST

100%

LAN DATA SOVEREIGNTY

$O(N+M)$

TRIMMED LCS DIFFING

ACID WAL

ZERO-LOSS MEMORY CORE

RUNTIME ARCHITECTURE

Native Cockpit

SwiftUI

60fps macOS deck. Real-time visual diff inspector, live Web Reflection viewer, and project artifact manager.

Automation Daemon

:7171 / :9222

Node.js Puppeteer stealth browser engine, screencast frame broker, tree-sitter AST indexer, and task leasing ring.

Dispatcher Gateway

:8000

Python 3.13 FastAPI gateway. Handles model proxying, DOM layout extraction, and task verification runner.

AUDIT & VERIFICATION LOOP

01 // DRAFT

Local Draft & Baseline

Local model creates draft; immutable baseline snapshot committed to memory ring.

02 // OFFLOAD

Headless Web Audit

Stealth browser offloads audit to DeepSeek or Kimi; live reflection streams to UI.

03 // DIFF

Trimmed LCS Diffing

Computes line-by-line mutation window in $O(N+M)$ with instant visual diffs.

04 // VERIFY

Closed-Loop Tests

Executes native linters, compilers, and test suites before final workspace commit.

ENGINE CAPABILITIES

Zero-Loss Version Control

Eliminates destructive overwrites. Pre-audit baselines and post-audit revisions are permanently linked with visual diff chains.

Live Web Reflection Window

Zero blind execution. Floating native viewer streams outbound prompts and inbound tokens from headless browser processes.

Semantic Context Compression

Distills lengthy chat history into executive semantic anchors while archiving literal verbose logs to disk.

Isolated Project Workspaces

All codebase operations execute within dedicated project directories. The Cockpit Artifacts drawer displays only active deliverables.

Status: All test suites passing (Swift package clean, 12/12 daemon, 8/8 dispatcher). STACK: Swift 5.9 · Node 20 · Python 3.13 · SQLite WAL